

ContraSynth: Amplifying Alignment Gradients via Counterfactual Flaw Synthesis

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Abstract. When aligning advanced text-to-image diffusion models, the efficacy of preference learning methods such as Direct Preference Optimization (DPO) faces a fundamental bottleneck: the weak gradient signals derived from positive-negative sample pairs in existing datasets. To address this, we introduce **ContraSynth**, a novel data-centric framework that shifts from passive sample selection to active counterfactual contrast synthesis. Our pipeline leverages a VLM-LLM synergy to synthesize negative samples by injecting specific, high-contrast flaws (including Semantic Divergence, Aesthetic Degradation, and Plausibility Violation) into high-quality source prompts. These hard negative samples force the model to distinguish between correct concepts and explicit failures, establishing high-gradient decision boundaries. By fine-tuning representative Stable Diffusion (SD) models, including SD1.5, SDXL, and SD 3.5 Large, ContraSynth significantly outperforms DPO baselines trained on standard preference datasets in prompt fidelity, aesthetic quality, and human preference, while structurally "unlearning" flawed concepts. Extensive human evaluation confirms a robust 62% preference rate over standard baselines. Crucially, as a purely data-centric enhancer, our method is orthogonal to optimization algorithms (*e.g.*, NPO) and reduces data preparation compute by approximately 75%.

Keywords: Data Synthesis · Preference Alignment · Image Generation

1 Introduction

The rapid evolution of text-to-image (T2I) diffusion models has revolutionized visual content creation, enabling the generation of remarkably realistic and diverse images from natural language descriptions [3, 9, 12]. Despite their impressive generative capabilities, ensuring that these models faithfully adhere to complex human intentions, maintain high aesthetic standards, and avoid generating implausible physical artifacts remains a critical challenge. To bridge this gap, preference alignment techniques, most notably Reinforcement Learning from Human Feedback (RLHF [6]) and Direct Preference Optimization (DPO [15]), have emerged as the *de facto* paradigm, successfully adapting alignment strategies from large language models to the visual domain.

However, while these optimization algorithms offer elegant formulations, their empirical efficacy is fundamentally constrained by the quality of the underlying

preference datasets, where the specific construction of positive and negative sample pairs (y_w, y_l) dictates the optimization landscape. Ideally, samples should be globally proximal yet significantly divergent in critical preference-related dimensions. Current datasets [5, 21] are typically constructed via rejection sampling, where the negative sample y_l is selected from a pool of model outputs based on a relatively low reward score. This process is stochastic and often yields merely "sub-optimal" pairs, which ultimately provides weak gradient signals and creates fuzzy decision boundaries during alignment.

To this end, we propose ContraSynth (Contrastive Synthesis), a framework that strengthens the alignment signal by actively synthesizing proximal counterfactual contrast samples. Instead of relying on passive natural flaws, this method proactively generates negative samples with targeted defects (*e.g.*, "a floating apple" for plausibility violation). By maintaining feature proximity while enforcing preference-critical divergence, the framework establishes a high-gradient decision boundary that facilitates robust and discriminative feature learning.

Our contributions are threefold:

1. We propose ContraSynth, a data-centric framework that shifts the T2I alignment paradigm from passive selection to active counterfactual synthesis, effectively resolving the weak gradient bottleneck in preference alignment.
2. We achieve state-of-the-art (SOTA) performance across multiple Stable Diffusion (SD) models (SD1.5, SDXL, and SD 3.5 Large). Crucially, we establish that ContraSynth acts as a universal data enhancer orthogonal to advanced algorithms (*e.g.*, NPO), while reducing data preparation compute by 75%.
3. We provide mechanistic insights using optimization dynamics and latent trajectory analysis, demonstrating that ContraSynth induces structural Concept Unlearning, enabling the model to robustly suppress erroneous outputs.

2 Related Work

Aligning Diffusion Models via Preference Learning. Aligning T2I models with human intent has evolved from simple supervised fine-tuning (SFT) to sophisticated preference learning paradigms. Early approaches relied on curated high-quality datasets, but these lacked the granularity to correct specific model failures. Inspired by Large Language Models (LLMs), RLHF was adapted for diffusion models [6]. This paradigm trains a reward model to proxy human judgment, using it to guide the generator.

However, the efficacy of RLHF is fundamentally bottlenecked by the *discriminative power* of the reward signal. As noted in recent distillation studies [20], pipeline optimization is often constrained by supervision signal quality. When the preference data exhibits high entropy (*i.e.*, ambiguous good/bad distinction), the resulting reward model provides a flat, noisy optimization landscape, making it difficult for the generator to escape local minima of "mediocrity".

The Data Bottleneck: From Selection to Synthesis. The foundation of effective alignment lies in the preference data itself. The community has invested

heavily in collecting large-scale human comparisons to train reward models like ImageReward [22] and HPSv2 [21]. These models significantly outperform traditional metrics (*e.g.*, FID, CLIP Score) in mimicking human aesthetic judgment.

However, a critical limitation persists in these selection-based datasets: the Weak Gradient Problem. In datasets constructed via ranking model outputs (*e.g.*, Pick-a-Pic), the negative sample y_l is often visually and semantically similar to the positive sample y_w , differing only in subtle details. This high similarity results in a fuzzy decision boundary, providing a weak or vanishing gradient signal during optimization. While synthetic data has shown promise in improving downstream tasks [1], most approaches focus on increasing data quantity or diversity rather than structural quality.

Optimization Algorithms vs. Signal Amplification. DPO [15] revolutionized alignment by deriving a surrogate objective to directly optimize the policy on preference pairs. Following this, a surge of algorithmic innovations aimed to squeeze more signal from existing data. Recent works have unified DPO with score matching for stability [24], proposed pairwise win-rates to mitigate reward hacking [18], and introduced step-by-step optimization to assess fine-grained details [7]. Notably, Negative Preference Optimization (NPO [16, 17]) aligns with our goal by explicitly focusing on what the model should *not* generate. However, NPO and similar methods [8] operate on fixed, pre-existing datasets. They attempt to amplify the learning signal purely through loss function engineering.

We argue that algorithmic refinement hits a ceiling imposed by the data’s intrinsic signal-to-noise ratio. Even the most sophisticated loss function cannot extract a strong gradient from a preference pair with negligible perceptual difference. Unlike algorithm-centric approaches [7, 16] that passively ingest weak negatives, or discussions on the privacy implications of synthetic data [2, 19], ContraSynth adopts a *data-centric* philosophy. By synthesizing Counterfactual Flaws, ContraSynth artificially amplifies the contrastive signal. This transforms the alignment process from discerning subtle nuances to learning from high-gradient decision boundaries, establishing a more robust principle for visual preference alignment.

3 Methodology

The core philosophy of ContraSynth is to replace the passive *selection* of weak negative samples with the active *synthesis* of high-signal counterfactuals. Our framework addresses the "Weak Gradient Problem" inherent in standard preference learning by artificially creating high-contrast decision boundaries. The pipeline comprises two tightly integrated stages:

1. *Contrastive Synthesis*: We leverage a VLM-LLM synergy to perform Counterfactual Attribute Injection, generating negative prompts explicitly designed to induce specific, controllable visual defects.
2. *High-Gradient Alignment*: The diffusion model trains on these synthesized contrastive pairs to sharpen its internal representations against targeted flaws.

Step 1. Contrastive Synthesis of Negative Samples

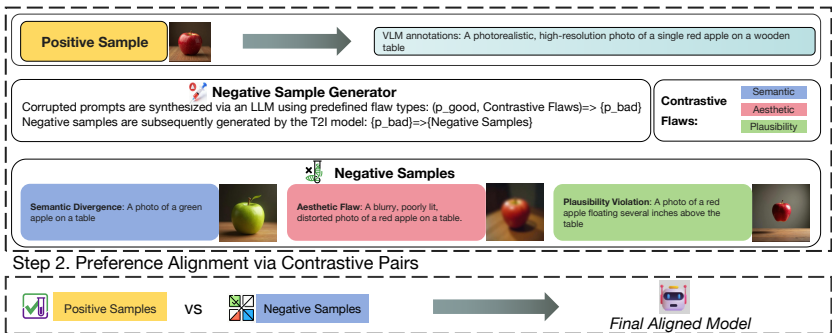


Fig. 1: An overview of ContraSynth. The process consists of two parts: 1) Contrastive Synthesis of Negative Samples. 2) Preference Alignment via Contrastive Pairs.

3.1 Taxonomy of Contrastive Flaws

To ensure the synthesized negatives provide meaningful learning signals rather than random noise, we categorize common T2I failure modes into a structured taxonomy (summarized in Table 1). This allows us to target specific weaknesses in the generative prior:

- **Semantic Divergence** (e_s): Fundamental misalignment between the core meaning of the prompt and the generated content. This includes incorrect object classes (e.g., a dog instead of a cat), attribute binding errors (e.g., a red car becoming blue), or missing key elements.
- **Aesthetic Degradation** (e_a): Failures in visual quality independent of semantic content. We target specific low-quality attributes such as poor lighting, blurriness, JPEG artifacts, and amateurish composition.
- **Plausibility Violation** (e_p): Breaches of real-world physics, anatomy, or logic. This critical category includes gravity violations (floating objects), geometric impossibilities (Escher-like structures), and anatomical distortions (e.g., malformed hands, extra limbs).

3.2 Counterfactual Flaw Synthesis

Our data generation process is fully automated, scalable, and designed to produce "Hard Negatives", *i.e.*, samples that are plausibly generated by the model but explicitly wrong. Let x_{src} be a high-quality source image (e.g., from a curated dataset or high-aesthetic subset).

Part 1: Structured Annotation. We employ a Vision Language Model (VLM) to generate a detailed, descriptive caption of x_{src} , which serves as the positive anchor prompt p_{good} . This ensures that p_{good} is highly aligned with the visual content of the positive sample.

Part 2: Counterfactual Attribute Injection. We introduce the Prompt-Injector, an LLM-based module that systematically perturbs p_{good} to create a

Table 1: Contrastive flaws in T2I and their corresponding prompt categories.

Contrastive Flaws	Descriptions	Prompt Categories where This Error Commonly Occurs
Semantic Divergence	Discrepancies between the prompt’s core semantics and the visual content of the generated image, such as incorrect subjects, attributes (<i>e.g.</i> , color, material), or actions.	Prompts requiring precise attribute binding (<i>e.g.</i> , "a red cube on a blue sphere"), complex object compositions, or detailed scene descriptions.
Aesthetic Flaw	Failures to meet established standards of artistic or photographic quality, such as flawed composition, poor lighting, unrealistic textures, or a failure to adhere to a specified artistic style.	Prompts requesting a specific artistic style (<i>e.g.</i> , "in the style of Van Gogh"), photographic techniques (<i>e.g.</i> , "a long-exposure photograph"), or high-quality, photorealistic outputs.
Plausibility Violation	Generation of content that contradicts real-world physics, anatomy, or commonsense logic, such as incorrect limb counts, impossible object interactions, or nonsensical scene contexts.	Prompts depicting humans and animals (especially hands), scenes with complex physical interactions (<i>e.g.</i> , reflections, shadows), and real-world scenarios.

negative prompt p_{bad} . Unlike random noise injection, the PromptInjector applies targeted transformations based on the flaw type $e \in \{e_s, e_a, e_p\}$. For a prompt $p_{\text{good}} = \text{"A photo of an apple on a table"}$, the injection logic follows:

- Plausibility e_p : Inject counter-physical descriptors (*e.g.*, "floating apple");
- Aesthetic e_a : Inject quality degradation tokens (*e.g.*, "blurry, out of focus");
- Semantic e_s : Swap key attributes (*e.g.*, "a green pear").

Part 3: Negative Synthesis. Finally, the target diffusion model $\mathcal{M}$ generates the negative sample x_{neg} conditioned on this perturbed prompt:

$$x_{\text{neg}} = \mathcal{M}(\text{PromptInjector}(p_{\text{good}}, e)) \quad (1)$$

This process yields a preference pair $(x_{\text{src}}, x_{\text{neg}})$ where the negative sample exhibits a specific, controlled failure while remaining otherwise semantically consistent with the positive sample. Crucially, because x_{neg} is generated by the model itself (guided by the bad prompt), it lies within the model’s feasible output space, making it a valid and informative training example.

3.3 Alignment via High-Gradient DPO

We employ Diffusion-DPO [15] to align the model. The objective maximizes the margin between the likelihood of the preferred data y_w (x_{src}) and the dispreferred data y_l (x_{neg}), both conditioned on the positive anchor prompt $c = p_{\text{good}}$:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta, \pi_{\text{ref}}) = -\mathbb{E}_{(c, y_w, y_l) \sim D} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | c)}{\pi_{\text{ref}}(y_w | c)} - \beta \log \frac{\pi_\theta(y_l | c)}{\pi_{\text{ref}}(y_l | c)} \right) \right], \quad (2)$$

The Gradient Amplification Effect. In standard DPO datasets, y_l is often similar to y_w in terms of preference dimensions (*e.g.*, a slightly inferior crop), leading to a small likelihood ratio difference and consequently, a vanishing gradient. In ContraSynth, y_l is explicitly synthesized to be a *counterfactual* opposite of y_w along a specific dimension (*e.g.*, gravity-compliant vs. gravity-defying). This maximizes the perceptual and semantic distance, resulting in a steeper loss surface around the decision boundary. This amplified signal forces the model to sharply distinguish between valid and invalid concepts, accelerating convergence and improving robustness.

3.4 Computational Efficiency Analysis

A key advantage of ContraSynth is its efficiency compared to standard rejection-sampling approaches (*e.g.*, Pick-a-Pic [5]). Standard construction requires generating K candidates (typically $K \geq 4$) and scoring them with a reward model $\mathcal{R}$ to filter a pair, with cost $C_{\text{mining}} \approx K \cdot C_{\text{gen}} + K \cdot C_{\text{reward}}$. In contrast, ContraSynth generates the negative sample directly via targeted prompting, requiring only one generation step:

$$C_{\text{synth}} \approx C_{\text{VLM}} + C_{\text{LLM}} + 1 \cdot C_{\text{gen}} \quad (3)$$

Since C_{VLM} and C_{LLM} (typically single-forward passes) are negligible compared to image generation, and $K \gg 1$, our method drastically reduces computational overhead. Empirically, for a dataset of $K = 4$, ContraSynth reduces total GPU hours by approximately **75%** compared to standard rejection sampling pipelines.

4 Experimental Validation

4.1 Experimental Design

Models and Baselines. To demonstrate the versatility of ContraSynth, we conduct experiments across three representative T2I backbones: SD 1.5 [12], SDXL [9], and SD 3.5 Large [3, 14]. For preference learning, we utilize the Pick-a-Pic [5] dataset as the source for positive samples. Our primary baseline is the standard DPO method trained on the original Pick-a-Pic dataset (which uses model-generated rejection sampling for negatives).

Data Synthesis. The ContraSynth pipeline is implemented using the Qwen2.5 family: Qwen2.5-VL-3B-Instruct [10] for annotation and Qwen2.5-7B-Instruct [23] as the PromptInjector. The specific prompt templates steering both the visual annotation and counterfactual injection stages are detailed in Appendix.

Evaluation Metrics. We employ a comprehensive suite of metrics covering three dimensions:

- **Prompt Fidelity:** CLIP Score [11] and VLM-CLIP Score (Qwen2.5-VL) to assess semantic alignment.
- **Aesthetic Quality:** LAION-Aesthetics (Laion Aes [13]) and Cafe-Aesthetics (Cafe Aes) for visual appeal.
- **Plausibility & Preference:** HPSv2 [21] and PickScore (PIS [5]), fine-tuned to align with human judgments on commonsense and overall quality.

4.2 Main Results: Quantitative Benchmarks

The main results, presented in Table 2, confirm that ContraSynth establishes a new state-of-the-art by achieving **comprehensive superiority** over the standard DPO baseline. Unlike prior methods that often trade off prompt fidelity for aesthetics, ContraSynth delivers consistent gains across *all* evaluated metrics and model architectures, validating the efficacy of synthesizing high-gradient counterfactuals.

Table 2: Main Results on T2I Generation Benchmarks. We compare ContraSynth against the base models and standard DPO baselines across three model scales. ContraSynth achieves consistent SOTA performance, validating the superiority of synthesizing high-gradient negative samples over passive selection.

Models	Methods	Prompt Fidelity		Aesthetic		Plausibility	
		CLIP	VLM CLIP	Laion Aes	Cafe Aes	HPSv2	PIS
SD 1.5	Base	25.65	80.21	5.14	0.734	0.241	20.24
	+ Standard DPO	25.87	81.15	5.19	0.736	0.246	20.44
	+ ContraSynth (Ours)	25.91	82.33	5.23	0.741	0.251	20.53
SD XL	Base	25.45	83.45	5.51	0.776	0.248	21.00
	+ Standard DPO	26.21	84.45	5.59	0.771	0.263	21.35
	+ ContraSynth (Ours)	26.34	85.36	5.71	0.792	0.281	21.37
SD 3.5 Large	Base	26.14	84.83	5.72	0.803	0.276	21.41
	+ Standard DPO	26.45	85.10	5.77	0.808	0.281	21.42
	+ ContraSynth (Ours)	26.71	85.52	5.81	0.824	0.283	21.45

Scalability and Universality. ContraSynth demonstrates robust improvements regardless of the model scale:

- On **SD 1.5**, we observe a uniform uplift across all six metrics, notably pushing the VLM-CLIP score from 81.15 to **82.33**, proving that our method can effectively extract remaining capacity even from mature, smaller architectures.
- On **SDXL**, the gains are substantial. We achieve top-tier performance not just in aesthetics (Cafe Aes: 0.771 $\rightarrow$ **0.792**) but also in semantic alignment (CLIP: 26.21 $\rightarrow$ **26.34**), significantly widening the gap with the baseline.
- Crucially, on **SD 3.5 Large**, where standard DPO begins to saturate (*e.g.*, PIS stagnating at 21.42), ContraSynth successfully breaks the performance ceiling, advancing to **21.45**. This indicates that even for highly optimized models, standard negatives provide insufficient gradient signals, whereas our targeted flaws expose and correct persistent, subtle weaknesses.

Structural Alignment with Flaw Taxonomy. The quantitative gains directly map to our design objectives. The simultaneous boost in HPSv2 & PickScore and CLIP & VLM-CLIP confirms that our model effectively learns to reject *Plausibility Violations* and *Semantic Divergence* respectively. By forcing the model to distinguish between correct concepts and explicitly synthesized failures, ContraSynth achieves a holistic alignment that standard DPO, limited by noisy, low-margin negatives, cannot replicate.

4.3 Qualitative Analysis: Visual Evidence of Flaw Correction

Experimental Setup. To validate the corrective power of ContraSynth beyond automated metrics, we conduct a side-by-side comparison on SDXL using four challenging prompts designed to probe specific failure modes: physical plausibility, anatomical correctness, semantic attribute binding, and aesthetic quality. Visual results are presented in Figure 2, comparing the Base model (Top), Standard DPO (Middle), and ContraSynth (Bottom).

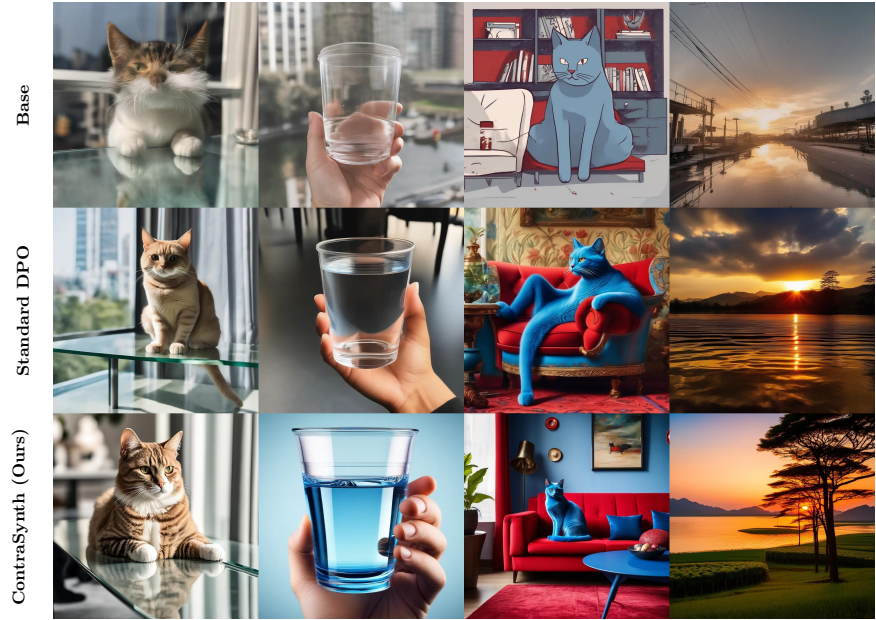


Fig. 2: Qualitative Comparison on SDXL. We compare samples generated by the Base model (Top), Standard DPO (Middle), and **ContraSynth (Bottom)**. **Columns 1-2 (Plausibility):** ContraSynth eliminates physical violations (e.g., tail penetrating glass, distorted hands) visible in baselines. **Column 3 (Semantic):** ContraSynth prevents color bleeding (blue cat, red sofa). **Column 4 (Aesthetic):** ContraSynth enhances lighting and composition.

Physical Logic & Anatomical Correctness (Columns 1 & 2). Standard DPO, while successfully improving texture, often fails to rectify fundamental physical hallucinations.

- **Physics (Col 1):** For the prompt "a cat sitting on a glass table", the Baseline exhibits severe logic violations, such as **a tail impossibly penetrating the glass surface**. ContraSynth correctly models the physical occlusion and surface interaction.
- **Anatomy (Col 2):** In the prompt "Holding a transparent cup filled with water", the Baseline struggles with biological constraints, rendering **distorted finger joints and impossible grip mechanics**. ContraSynth respects anatomical priors, generating a geometrically accurate hand with correct transparency refraction.

Semantic Attribute Binding (Column 3). We observe a critical advantage in disentanglement. For the prompt "The blue cat is sitting on the red sofa", the Baseline suffers from **color pollution**, where the "blue" attribute bleeds into the "red" sofa (rendering it purple). ContraSynth achieves precise attribute binding, maintaining a vivid blue cat against a red background without semantic leakage.

Aesthetic Quality (Column 4). Finally, regarding artistic rendering, we test the prompt "The scenery under the setting sun". The Baseline produces a flat, visually mundane image with limited dynamic range. In contrast, ContraSynth significantly elevates the aesthetic appeal, synthesizing a cinematic scene with **dramatic lighting, rich color gradients, and superior composition.**

Conclusion. These comparisons confirm that ContraSynth does not merely optimize a score; it fundamentally sharpens the model’s decision boundaries. By training on synthesized counterfactuals (*e.g.*, explicit gravity violations or color mismatches), the model effectively learns to suppress specific artifacts, resulting in generations that are physically robust, semantically accurate, and aesthetically superior.

4.4 Impact of Optimization Algorithms

Data-Centric vs. Algorithm-Centric. A natural question is whether the gains stem from the data quality or the choice of optimization objective. To address this, and to demonstrate the universality of ContraSynth, we compare our data-centric approach against advanced algorithmic baselines, specifically NPO [16].

Experimental Setup. We conduct a cross-comparison on SDXL using two datasets (Standard Pick-a-Pic vs. ContraSynth) and two algorithms (DPO vs. NPO).

Analysis. As shown in Table 3, while switching from DPO to NPO on standard data yields a marginal improvement (+0.005 HPSv2), switching to ContraSynth data with the standard DPO yields a much larger gain (+0.018 HPSv2). This supports our claim that **data quality is the primary bottleneck**. Furthermore, combining ContraSynth with NPO achieves the highest performance, demonstrating that our high-gradient counterfactuals are **orthogonal** to algorithmic improvements and can serve as a universal enhancer for preference learning frameworks.

Table 3: Data vs. Algorithm. Comparing the impact of improving the *algorithm* (DPO → NPO) versus improving the *data* (Standard → ContraSynth). Results on SDXL show that enhancing data quality yields larger gains than changing the algorithm, and ContraSynth boosts NPO further.

Dataset	Algorithm	HPSv2	PickScore	Gain Source
Standard	DPO (Base)	0.263	21.35	-
Standard	NPO	0.268	21.35	Algorithm Only
Ours	DPO	0.281	21.37	Data Only
Ours	NPO	0.284	21.41	Data + Algo

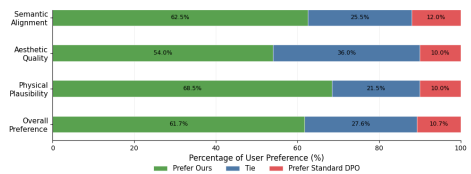


Fig. 3: Human Preference Evaluation. ContraSynth consistently outperforms Standard DPO, with the most significant gains in *Physical Plausibility* (68.5%) and *Semantic Alignment* (62.5%), confirming the efficacy of targeted counterfactual synthesis.

4.5 Human Preference Evaluation

Given the limitations of automated metrics, we conducted a blind human user study to provide the definitive verdict on generation quality. We recruited 20 participants to evaluate 100 image pairs (Standard DPO vs. ContraSynth) across our three flaw categories.

Results. Figure 3 illustrates the win rates. ContraSynth achieves a decisive victory with an overall win rate of **61.7%** (vs. 10.7% for DPO). Crucially, the margin is widest in **Physical Plausibility** (**68.5%** win rate). This empirically validates that standard datasets, which lack explicit negative signals for physical errors, fail to align models with human common sense. By proactively synthesizing these failures, ContraSynth aligns the model with the nuanced physical reality preferred by human observers.

5 In-depth Analysis and Discussion

To elucidate the internal mechanisms of ContraSynth beyond aggregate metrics, we conduct a series of investigative studies. Our analysis begins with a Mechanism Analysis (Section 5.1), where we employ cross-attention maps and latent trajectory visualizations to verify that ContraSynth induces genuine structural rectification within the diffusion process. In Section 5.2, we provide Empirical Evidence of Signal Amplification by monitoring optimization dynamics, demonstrating how synthesized negatives effectively resolve the vanishing gradient problem and sharpen the loss landscape. Finally, we present an Ablation Study (Section 5.3) on various flaw types and mixing schemes to quantify their individual and synergistic contributions to the final alignment.

5.1 Mechanism Analysis: Validating Concept Unlearning

To investigate the underlying mechanism of robust generalization, we analyze the internal representations of the diffusion model, specifically focusing on cross-attention maps and latent probability densities. We hypothesize that training on counterfactual flaws actively induces *Concept Unlearning*, penalizing the activation of specific failure modes in the latent space even when error-inducing keywords are absent.

Cross-Attention Map Visualization. We employ Diffusion Attentive Attribution Maps (DAAM) to visualize the cross-attention weights between spatial features and specific text tokens during the denoising process. Figure 4 (Left) compares the attention maps of the Base SDXL and ContraSynth. When given a clean prompt ("clean" refers to the unperturbed prompt, devoid of any explicit positive or negative prompts intended for error avoidance, *e.g.*, "an apple on a table"), the Base model often exhibits residual attention towards implausible regions (*e.g.*, empty space above the table, leading to a "floating" artifact). In contrast, ContraSynth significantly suppresses the attention weights in these non-physical regions, sharply localizing the object to valid supporting surfaces. This confirms that the model has unlearned the spatial prior of "floating" objects.

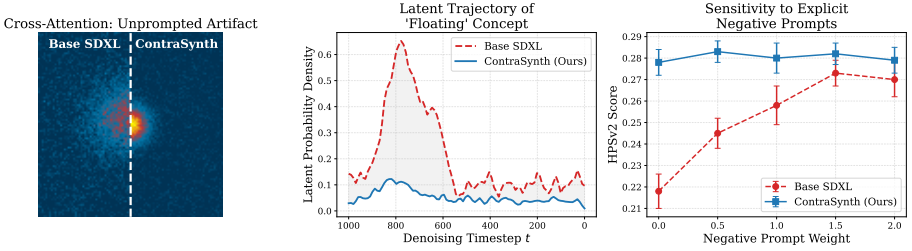


Fig. 4: Mechanism of Concept Unlearning. *Left:* DAAM cross-attention visualization for an unprompted implicit concept. ContraSynth sharply suppresses spurious attention in implausible regions. *Middle:* The latent projection trajectory of the "floating" concept across denoising timesteps. ContraSynth fundamentally inhibits the probability accumulation of the flaw. *Right:* Model sensitivity to Negative Prompt (NP) weights. ContraSynth achieves peak HPSv2 without relying on heavy NP guidance, proving the flaws are structurally unlearned rather than merely suppressed by CFG.

Latent Probability Trajectory. To further quantify this suppression, we monitor the probability density of implicit flawed concepts during inference. Although these negative tokens (*e.g.*, "blurry", "floating") are not present in the clean prompt, we compute their projection scores against the intermediate latent representations z_t across diffusion timesteps t . As illustrated in Figure 4 (Middle), the Base model exhibits a gradual accumulation of probability mass for the "floating" concept during the early structure-formation steps ($t \in [800, 1000]$). Conversely, ContraSynth maintains a strictly suppressed probability trajectory throughout the entire denoising process, empirically validating that the failure mode has been systematically excised from the generative manifold.

Sensitivity to Negative Prompts. Finally, we evaluate the model's sensitivity to explicit negative prompts. Figure 4 (Right) shows the generation quality (HPSv2) as a function of the negative prompt weight. ContraSynth achieves optimal performance even with an empty negative prompt (weight = 0.0), whereas the Base model relies heavily on high negative prompt weights to achieve comparable quality. This diminished reliance demonstrates that ContraSynth internalizes the negative constraints into its core weights, shifting from passive suppression via Classifier-Free Guidance (CFG) to active structural unlearning.

5.2 Optimization Dynamics: Empirical Proof of Signal Amplification

A core theoretical claim of ContraSynth is that standard alignment datasets suffer from the "Weak Gradient Problem", leading to fuzzy decision boundaries. By injecting counterfactual flaws, ContraSynth aims to artificially amplify the alignment signal. To empirically validate this, we analyze the optimization dynamics during the DPO fine-tuning of SDXL. The results are presented in Figure 5.

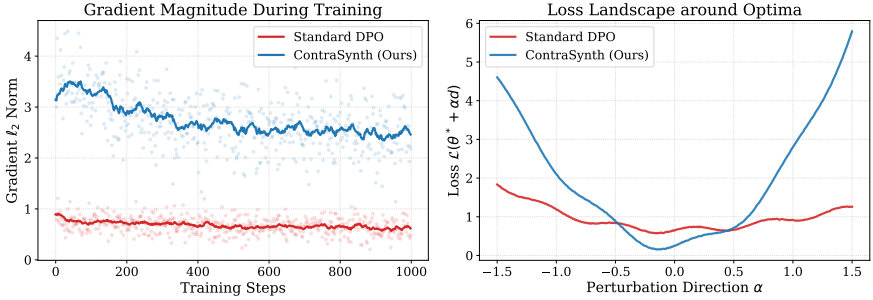


Fig. 5: Optimization Dynamics. *Left:* The ℓ_2 norm of the gradient during training (scatter plots denote batch-level gradients; solid lines denote exponential moving averages). ContraSynth yields a substantially stronger and more decisive optimization signal compared to Standard DPO. *Right:* 1D loss landscape visualization around the optima. ContraSynth creates a sharper decision boundary (a steep, deep minima) compared to the flat, rugged landscape of the baseline.

Gradient Magnitude and Variance. We monitor the ℓ_2 norm of the gradients with respect to the model parameters throughout the training steps. As shown in Figure 5 (Left), Standard DPO exhibits a consistently low gradient norm with minimal variance. Because the visual and semantic differences between standard y_w and y_l are subtle, the likelihood ratio difference remains marginal, yielding a faint optimization signal. In stark contrast, ContraSynth yields a significantly larger initial gradient norm accompanied by high batch-level variance. By maximizing the perceptual distance via explicit hard negatives, our method forces a robust policy update, effectively solving the vanishing gradient issue before stabilizing as the model converges.

Sharpening the Loss Landscape. The magnitude of the gradient directly influences the geometry of the optimization landscape. In Figure 5 (Right), we visualize the 1D loss landscape around the converged optima θ^* by perturbing the parameters along a principal direction d scaled by α . As observed, Standard DPO results in a remarkably flat, rugged minima prone to local sub-optima, corroborating our hypothesis of a "fuzzy decision boundary". ContraSynth, however, carves a sharp, steep basin. This geometric property confirms that synthesizing explicit counterfactuals establishes high-contrast exclusion zones, leaving no ambiguity about what constitutes a failure mode.

5.3 Ablation Study on Flaw Types and Mixing Schemes

Having validated the quality of our synthetic data and its impact on optimization, we now investigate how different contrastive flaws contribute to the final alignment. Table 4 compares models trained on single-error datasets (IDs 3-5) versus mixed strategies (IDs 6-10).

Table 4: Ablation study on contrastive flaw types. "Multi" (ID 10) represents our full method. The results show that targeting specific flaws improves corresponding metrics, and mixing them yields the most robust overall performance.

ID	Error Type	Prompt Fidelity		Aesthetic		Plausibility	
		CLIP Score	VLM CLIP	Laion Aes	Cafe Aes	HPSv2	PickScore
1	None	25.45	83.45	5.507	0.776	0.248	21.00
2	Untargeted	25.15	82.75	5.651	0.635	0.261	20.92
3	Sem	25.58	84.92	<u>5.792</u>	0.661	0.275	21.28
4	Aes	25.62	85.31	5.801	0.788	0.274	21.31
5	Plau	23.41	64.51	5.365	0.415	0.251	21.01
6	3+4	<u>25.85</u>	85.70	5.785	0.765	<u>0.279</u>	21.45
7	3+5	25.28	83.60	5.768	0.651	0.271	21.12
8	4+5	24.91	81.92	5.682	0.645	0.258	20.75
9	3+4+5	25.05	81.68	5.645	0.601	0.260	20.82
10	Multi	26.34	<u>85.36</u>	5.710	0.792	0.281	<u>21.37</u>

Targeted Gradients Sharpen Specific Capabilities. The single-type results (IDs 3-5) confirm that training on specific counterfactuals directly improves the corresponding metric.

- **Semantic (ID 3):** Training on semantic divergence yields the highest VLM-CLIP score improvement among single types, proving that explicit negative examples help the model bind attributes correctly.
- **Aesthetic (ID 4):** This configuration achieves the highest Laion-Aes score (5.801), demonstrating that penalizing specific visual artifacts (blur, noise) is more effective than generic tuning.
- **Plausibility (ID 5):** Directly targeting physics violations is critical to suppressing "hallucinations" like floating objects. However, doing so in isolation causes a notable drop in aesthetic and semantic metrics (*e.g.*, Cafe Aes drops to 0.415), indicating catastrophic forgetting when over-optimizing a single complex dimension.

Synergy via Multi-Flaw Counterfactuals. The most significant finding lies in our mixing schemes. We design two distinct strategies to combine flaws: **Mixed Datasets** (IDs 6-9), where datasets are formed by pooling single-flaw samples in equal proportions (*e.g.*, ID 9 contains 1/3 semantic-only, 1/3 aesthetic-only, and 1/3 plausibility-only negatives), and **Multi-flaw Samples** (ID 10, Multi), where the prompt for every single negative example is modified to jointly induce semantic, aesthetic, and plausibility errors simultaneously.

Interestingly, the naive dataset mixture of all three single-flaw pools (ID 9: 3+4+5) significantly degrades overall performance (*e.g.*, CLIP drops to 25.05). We hypothesize that this is caused by *gradient interference* across samples: penalizing a negative image that is physically flawed but aesthetically pristine can unintentionally degrade the model's robust aesthetic prior, causing conflicting optimization directions across the dataset.

In stark contrast, our **Multi** configuration (ID 10) achieves the best overall synergy and state-of-the-art balance (CLIP: 26.34, HPSv2: 0.281). By engineering a single, comprehensively flawed training sample, we maximize the semantic and perceptual margin $\Delta(y_w, y_l)$ between the positive anchor and the negative counterfactual. This unified, unambiguous negative signal prevents the model from exploiting optimization shortcuts, forcing it to establish a steep, high-gradient decision boundary across all three dimensions simultaneously without inter-task conflict.

Conclusion. The ablation study validates our core thesis: reliance on untar- geted or weak negative samples (ID 2) yields suboptimal results. Furthermore, merely mixing single-flaw datasets is insufficient due to competing gradients. The deliberate synthesis of Multi-Flaw Counterfactuals provides the most po- tent, unambiguous gradient signal, driving comprehensive model alignment and effectively regularizing the generative prior.

6 Conclusions and Limitations

Conclusion. This work establishes that the bottleneck in robust visual model alignment lies not in the complexity of optimization algorithms, but in the *signal-to-noise ratio* of the training data. The alignment of advanced T2I models is fundamentally constrained by the *Weak Gradient Problem* inherent in conven- tional datasets, where the perceptual difference between positive and negative samples is often too subtle to drive a decisive policy update.

ContraSynth introduces a paradigm shift from passive data selection to Ac- tive Counterfactual Synthesis. Instead of searching for "sub-optimal" samples, ContraSynth proactively engineers "Hard Negatives" by injecting specific, high- contrast flaws (Semantic Divergence, Plausibility Violation, and Aesthetic Degrada- tion). By training on these synthesized contrastive pairs, we establish *High- Gradient Decision Boundaries* that force the model to explicitly distinguish be- tween valid concepts and targeted failure modes.

Our extensive evaluations on SDXL and SD 3.5 Large demonstrate that this data-centric approach yields consistent SOTA performance.

Limitations. While ContraSynth demonstrates a potent new direction, we ac- knowledge several avenues for future research:

1. *Granularity of Flaw Taxonomy:* While the current work focuses on three fundamental error categories, expanding the number of categories could en- hance capabilities. However, generalization to natural, difficult-to-describe- with-prompts error categories remains an open research question.
2. *Prompt Injection Fidelity:* The precision of our counterfactuals depends on the underlying LLM's understanding of visual nuances. As multimodal LLMs evolve, we anticipate even more precise control over attribute injection and disentanglement.